

Q1 Choose the correct answer.

1 mark for each

1. How many bits are needed to store one BCD digit?
 A. 2 bits
 C. 3 bits

☒ B. 4 bits
 D. 1 bit

2. Convert $(312)_8$ into decimal.
 A. $(201)_{10}$
 C. $(203)_{10}$

B. $(202)_{10}$
 D. $(204)_{10}$

$$312_8 \Rightarrow \text{dec} \quad \frac{3 \times 8^2 + 1 \times 8^1 + 2 \times 8^0}{8} = 202$$

$$\frac{312}{8} = 39$$

$$39/8 = 4$$

$$4/8 = 0$$

3. What is the addition of the binary number $101001 + 010011 = ?$
 A. 010100
 C. 000111

☒ B. 111100
 D. 101110

$$\begin{array}{r} 101001 \\ + 010011 \\ \hline 111100 \end{array}$$

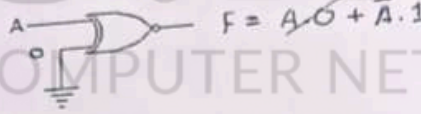
4. The following hexadecimal number $(1E,43)_{16}$ is equivalent to
 A. $(36.506)_8$
 C. $(35.506)_8$

B. $(36.206)_8$
 D. $(35.206)_8$

A	10
B	11
C	12
D	13
E	14
F	15

$$(1E,43)_{16} \Rightarrow (1 \times 16^1 + 14 \times 16^0, 4 \times 16^1 + 3 \times 16^0) = (30, 67)$$

$$XY + \bar{X}\bar{Y} = X \oplus Y$$



5. For the gate in the given figure the output will be
 A. 0
☒ C. A'

B. 1
 D. A

$$1000$$

6. What is the binary equivalent of the decimal number $(368)_{10} \Rightarrow ()_2$
☒ A. 101110000
 C. 111010000

B. 110110000
 D. 111100000

$$\frac{368}{2} = 184$$

$$\frac{184}{2} = 92$$

$$\frac{92}{2} = 46$$

$$\frac{46}{2} = 23$$

$$\frac{23}{2} = 11$$

$$\frac{11}{2} = 5$$

$$\frac{5}{2} = 2$$

$$\frac{2}{2} = 1$$

7. The simplification of the Boolean expression is.

$$\begin{aligned} F &= \overline{A}B\bar{C} + \overline{A}\bar{B}C \\ &= A + \bar{B} + C + \bar{A} + B + \bar{C} \\ &= (A + \bar{A}) + (\bar{B} + B) + (C + \bar{C}) \\ &= 1 + 1 + 1 \\ &= 1 \end{aligned}$$

A. A

☒ C. 1

B. 0

D. BC

0 1 1 0

0	1	1	0
0	1	1	0
0	1	1	0
0	1	1	0

$$\bar{A}BCD = 6$$

$$0110 =$$

8. The Gray code for decimal number 6 is equivalent to
- A. 1100
B. 1001
C. 0101
D. 0110

9. The Boolean expression $\bar{A}.B + A.\bar{B} + A.B$ is equivalent to.

- A. $A+B$
B. $\bar{A}.B$
C. $\overline{A+B} = \bar{A} + \bar{B}$
D. $A.B$
- $\bar{A} + \bar{B} = \overline{A.B}$
 $\overline{Y.Z} = \overline{Y.Z} = (\overline{Y.Z})(X.Y)$

10. the complement of $f = x' + yz' + xy'z$ is:
- A. $x' + (y' + z)(x' + y + z')$
B. $x'(y + z')(x + y' + z)$
C. $x(y' + z)(x' + y + z')$
D. $x + y'z + x'yz'$
- $F = \bar{x} + y\bar{z} + x\bar{y}z$
 $\bar{F} = x + \overline{y\bar{z}} + \overline{x\bar{y}z}$
 $\bar{F} = x + (\bar{y} + z)(\bar{x} + y + \bar{z})$
 $1 + 1 + 1 = 1$

11. The logic circuit shown can be minimized to

A. X — NOT — $\bar{X}$

B. X — AND — Y

C. X — AND — Y

D. X — ?

$\bar{X} \oplus \bar{Y} = \bar{X}\bar{Y} + X\bar{Y}$
 $= (\bar{X}\bar{Y})(\bar{X}\bar{Y})$
 $= (X+Y)(\bar{X}+\bar{Y})$
 $\bar{A} = X\bar{Y} + \bar{X}Y$
 $\bar{X}Y + XY$
 $Y(X+\bar{X})$
 Y

$\bar{X}\bar{Y} + X\bar{Y} + \bar{X}Y$
 $\bar{X}(\bar{Y}+1) + X\bar{Y}$
 $\bar{X} + X\bar{Y}$

12. The output of a logic gate is 1 when all its inputs are at logic 0. the gate is either

- A. a NAND or an EX-OR
B. an OR or an EX-NOR
C. an AND or an EX-OR
D. a NOR or an EX-NOR

A	B	F
0	0	1
0	1	0
1	0	0
1	1	0

Fill the answers for Q1 here

1	2	3	4	5	6	7	8	9	10	11	12
B	C	B	C	C	A	C	D	C	C	C	A

* Q2) Perform the subtraction on the given numbers using the 2's complement: (3 marks)

$$(1110)_2 - (10001)_2$$

$$2's \text{ comp } (10001) = 10001$$

$$(1110) + 10001$$

$$2's \text{ comp} = 1111$$

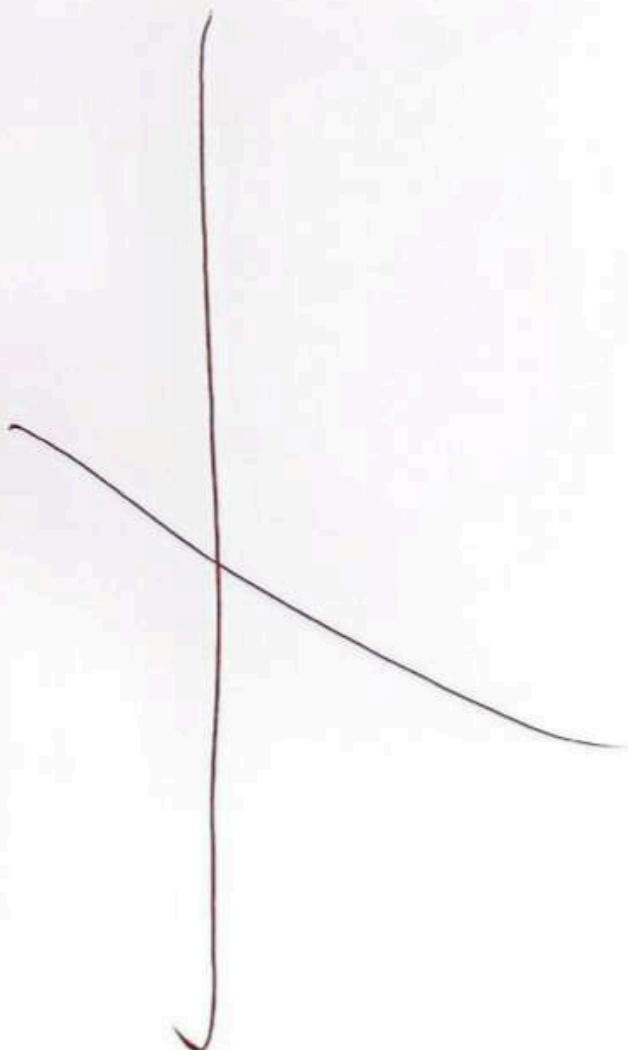
$$\begin{array}{r} 1110 \\ + 1111 \\ \hline 11101 \end{array}$$

$$1101 = (1110)_2 - (10001)_2$$



Q3) Derive the circuits for a three-bit parity generator using an odd parity bit with showing the truth table for the parity generator. (3 marks)

?



Q4) Simplify and draw the logic diagram for the given expression. (4 marks)

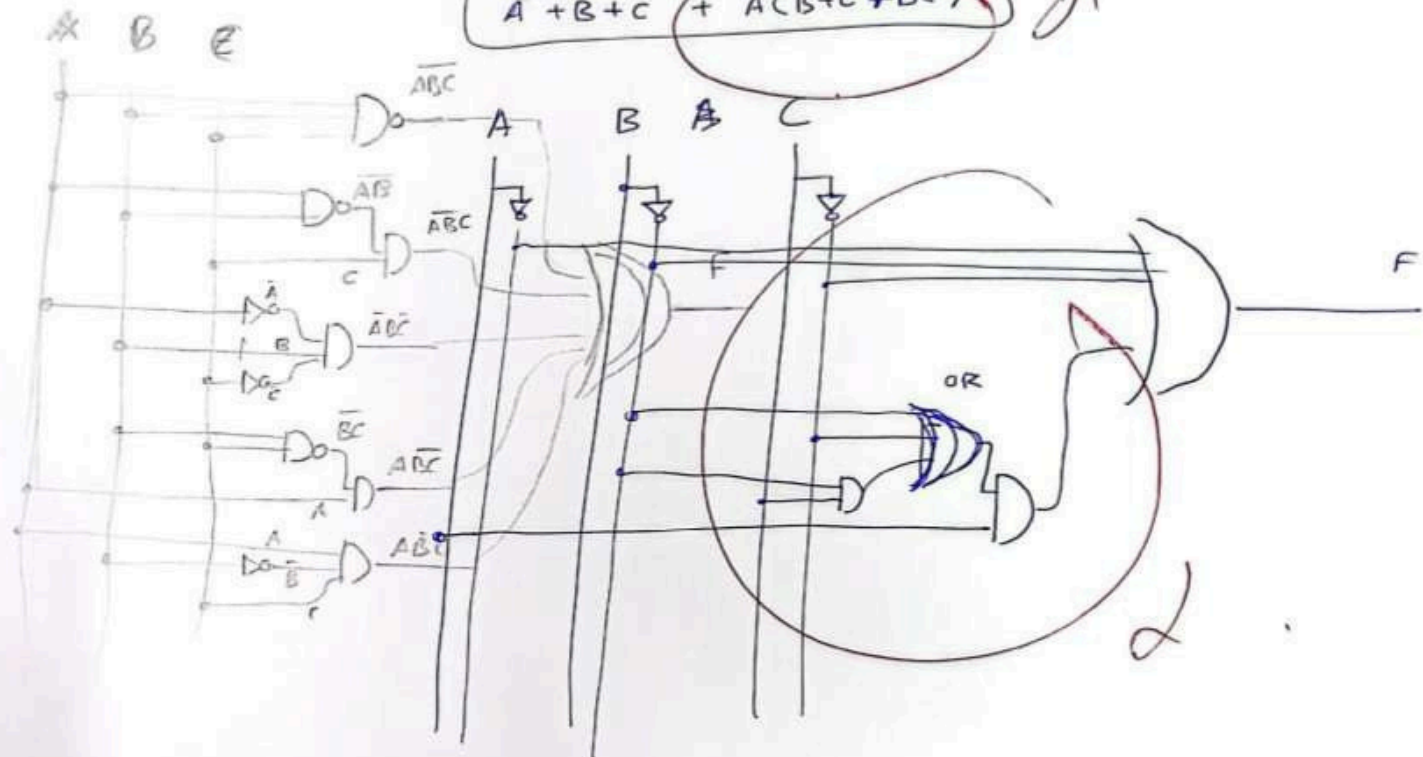
$$F = \overline{A}BC + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C}$$

$$F = \overline{A} + \overline{B} + \overline{C} + (\overline{A} + \overline{B})C + \overline{A}B\overline{C} + A\overline{B} + A\overline{C} + A\overline{B}C$$

$$\overline{A} + \overline{B} + \overline{C} + \overline{A}C + \overline{B}C + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C}$$

$$\overline{A}(1+C) + \overline{B}(1+C) + \overline{C}(1+\overline{A}B) + \overline{A}B + \overline{A}C + \overline{A}B\overline{C}$$

$$\overline{A} + \overline{B} + \overline{C} + A\overline{B} + \overline{C} + \overline{B}C$$



$$F = \overline{A}\overline{B} + C + A\overline{C}\overline{D}$$

Q5) For the given Boolean function (8 marks)

$$F(A,B,C,D) = \overline{A}\overline{B}\overline{C} + A\overline{C}\overline{D} + BC$$

Which has the don't care condition

$$d(A,B,C,D) = \sum(2,3,4,10,11)$$

a. Draw the K-map for function f (2marks)

b. Minimize f as product-of-sum (without don't care) (2marks)

c. Minimize f as sum-of-products (with don't care) (2marks)

$$F = \prod M = (5, 9, 12) = \overline{A}\overline{B}\overline{C}D$$

$$F = \sum m =$$

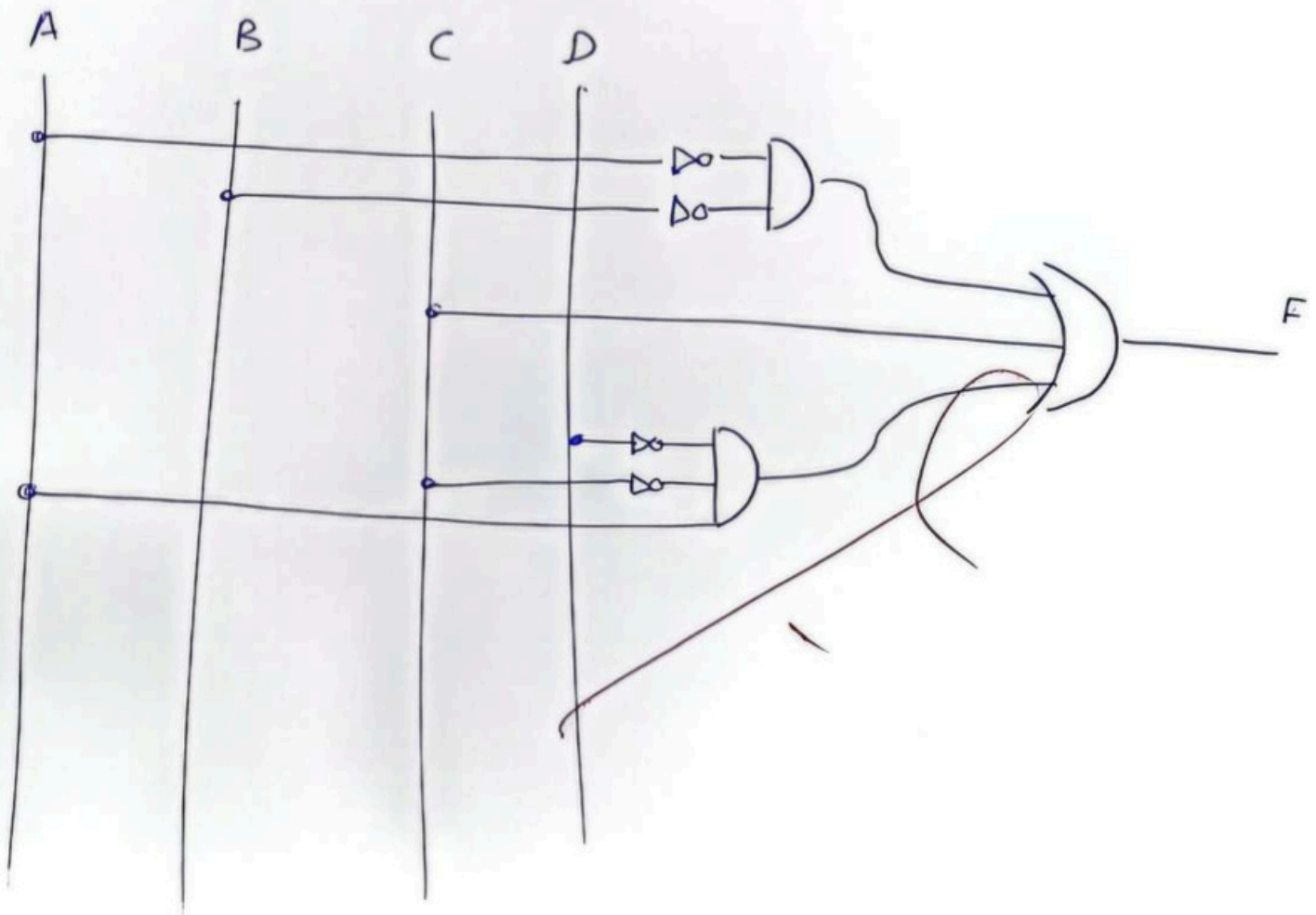
$$F = \overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}\overline{C}D + \overline{A}\overline{B}C\overline{D} + \overline{A}\overline{B}CD + \overline{A}B\overline{C}\overline{D} + \overline{A}B\overline{C}D + \overline{A}BC\overline{D} + \overline{A}BCD$$

W

	$\overline{C}\overline{D}$	$\overline{C}D$	CD	$C\overline{D}$
$\overline{A}\overline{B}$	1	1	X	X
$\overline{A}B$	X	0	1	1
AB	1	0	1	1
$A\overline{B}$	1	0	X	X

d. convert and draw the minimized f in branch (c) to NAND gate only circuit. (2 marks)

$$F = \bar{A}\bar{B} + C + A\bar{C}\bar{D}$$



Q.1)

1	2	3	4	5	6	7	8	9	10	11	12
B	B	B	B	C	A	C	C	A	C	B	D

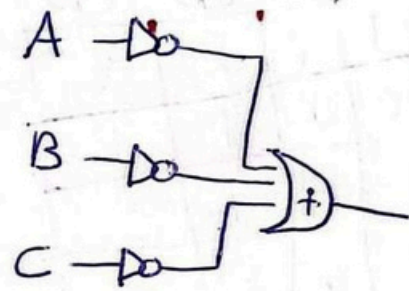
Q.2) $= -(00011)_2 = -3$

Q.3)

z	y	x	P
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

$$P = (x \oplus y \oplus z)'$$

Q.4) $f = \bar{A} + \bar{B} + \bar{C}$



Q. 5)

a)

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1	1	X	X
$\bar{A}B$	X	0	1	1
AB	1	0	1	1
$A\bar{B}$	1	0	X	X

b)
$$F = (\bar{B} + C + \bar{D})(\bar{A} + C + \bar{D})$$

c)
$$F = \bar{A}\bar{B} + C + A\bar{D}$$